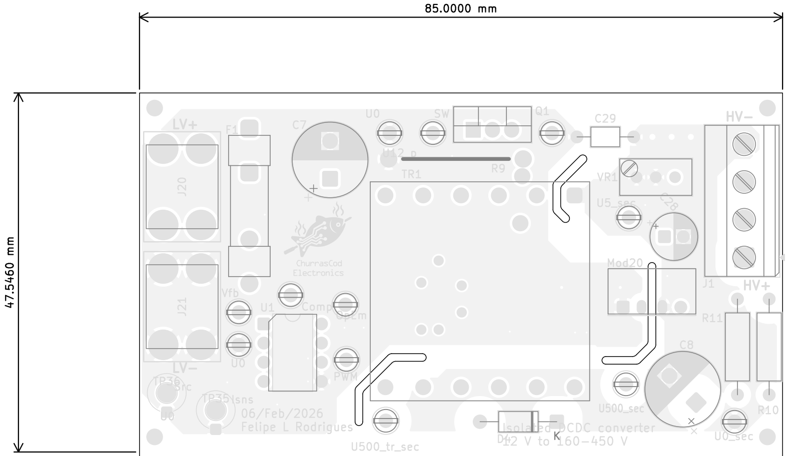
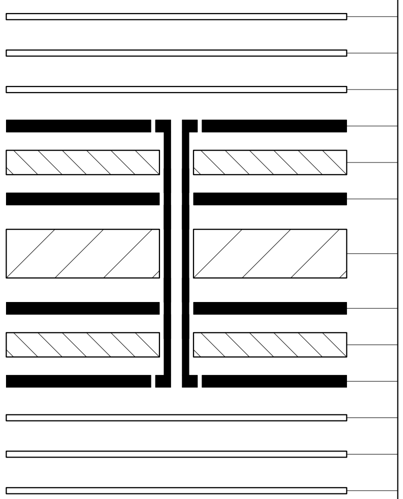
# Step-Up Fabrication Document

## Top Fabrication (Scale 1:1)




All dimensions are in millimeters unless otherwise specified.

### Layer Stack Legend



| Material | Layer | Thickness | Dielectric | Type | Gerber |
|---|---|---|---|---|---|
| | F.Paste | | | Paste Mask | |
| | F.Silkscreen | | Direct Printing | Legend | GBR |
| | F.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| **Copper** | **L1 (Ref)** | **0.07mm (2.00oz)** | | **Plane** | **GBR** |
| Prepreg | | 0.001mm | FR4_7628 | Dielectric | |
| **Copper** | **Virtual (Top side jumpers)** | **0.035mm (1.00oz)** | | **Copper Layer** | **GBR** |
| Core | | 1.35mm | FR4 | Dielectric | |
| **Copper** | **Virtual (Bot side jumpers)** | **0.035mm (1.00oz)** | | **Copper Layer** | **GBR** |
| Prepreg | | 0.001mm | FR4 | Dielectric | |
| **Copper** | **L2 (Sig, PWR)** | **0.07mm (2.00oz)** | | **Signal** | **GBR** |
| | B.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| | B.Silkscreen | | Direct Printing | Legend | GBR |
| | B.Paste | | | Paste Mask | |

Total thickness: 1.602mm
Note: external layer thicknesses are specified after plating

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
4) SURFACE FINISH: IMMERSION GOLD
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:
   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

| | |
|---|---|
| BOARD SIZE | 85.000 × 100.000 mm |
| BOARD THICKNESS | 1.602 mm |
| TRACE WIDTH | 0.300 mm |
| TRACE TO TRACE | -0.000 mm |
| MIN. HOLE (PTH) | 0.250 mm |
| MIN. HOLE (NPTH) | 2.100 mm |
| ANNULAR RING | 0.150 mm |
| COPPER TO HOLE | 0.200 mm |
| COPPER TO EDGE | 0.250 mm |
| HOLE TO HOLE | 0.254 mm |

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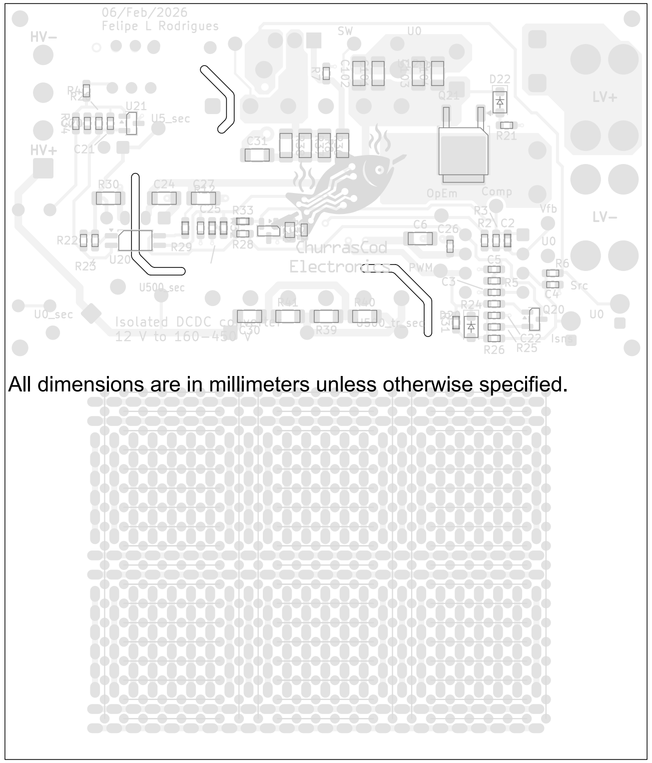
|---|---|---|---|---|
| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: **Step-Up** | | Project Name: **Isolated step-up 12:450 V** | |
| Sheet Title: Top Fabrication (Scale 1:1) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: **A4** | Sheet: **1** of **7** |

# Step-Up Fabrication Document

## Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

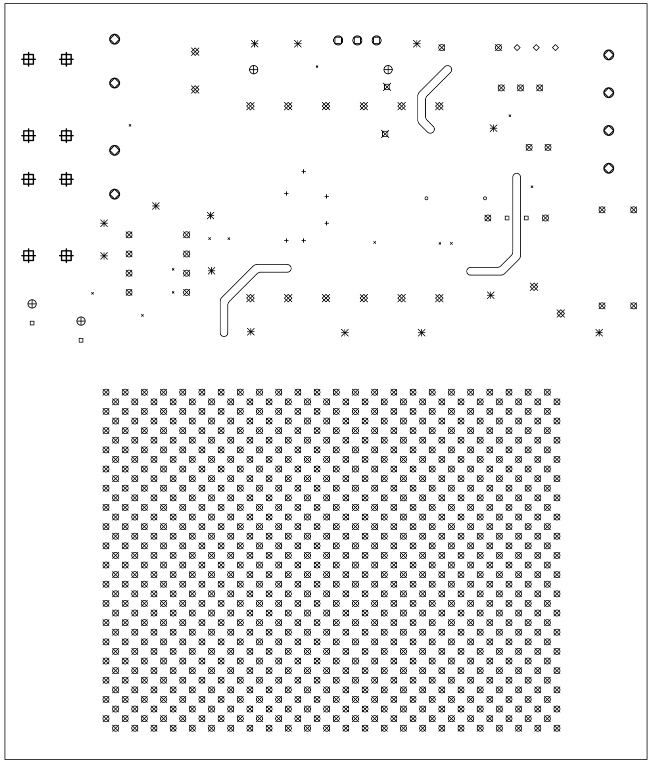
| | | | | |
|---|---|---|---|---|
| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: **Step-Up** | | Project Name: **Isolated step-up 12:450 V** | |
| Sheet Title: Bottom Fabrication (Scale 1:1) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: A4 | Sheet: 2 of 7 |

# Step-Up Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

## Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 13 | 0.25mm (9.84mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Via |
| ○ | 2 | 0.40mm (15.75mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Via |
| + | 6 | 0.50mm (19.69mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Via |
| □ | 4 | 0.70mm (27.56mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ◇ | 3 | 0.80mm (31.50mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Via |
| ⦻ | 885 | 0.80mm (31.50mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ✳ | 14 | 0.90mm (35.43mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ⌧ | 2 | 1.00mm (39.37mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Via |
| ※ | 16 | 1.00mm (39.37mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ⊕ | 4 | 1.10mm (43.31mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ▢ | 3 | 1.20mm (47.24mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ◈ | 8 | 1.30mm (51.18mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| ⊞ | 8 | 1.80mm (70.87mils) | PTH | Round | L1 (Ref) - L2 (Sig, PWR) | Pad |
| | Total 968 | | | | | |



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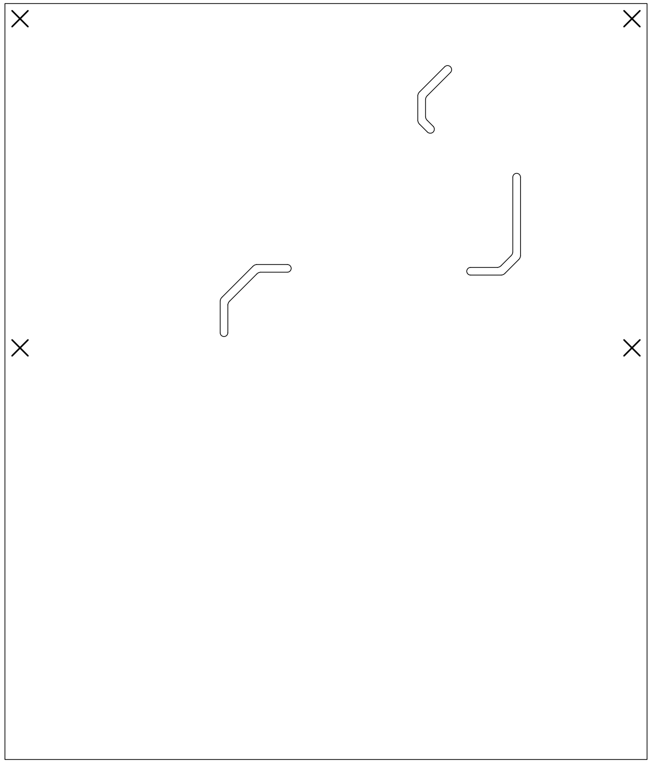
| | | | | |
|---|---|---|---|---|
| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: **Step-Up** | | Project Name: **Isolated step-up 12:450 V** | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: **A4** | Sheet: **3** of **7** |

# Step-Up Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

### Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 4 | 2.10mm (82.68mils) | NPTH | Round | L1 (Ref) - L2 (Sig, PWR) | Mechanical |
| | Total 4 | | | | | |



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| | | | | |
|---|---|---|---|---|
| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: Step-Up | | Project Name: Isolated step-up 12:450 V | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: A4 | Sheet: 4 of 7 |

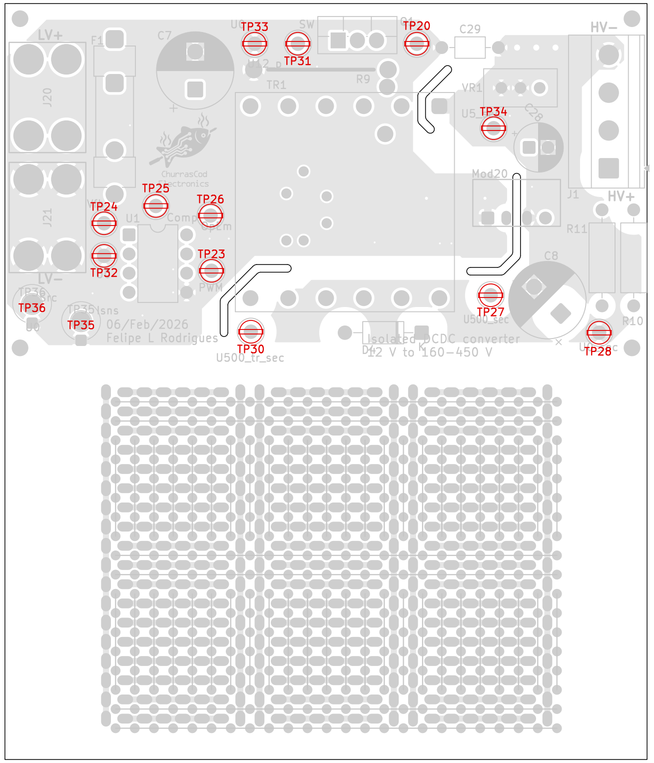
# Step-Up Fabrication Document

## Top Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|
| TP20 | SW | 56.53 | 44.66 |
| TP23 | PWM | 29.35 | 14.62 |
| TP24 | Vfb | 15.13 | 20.91 |
| TP25 | Comp | 21.98 | 23.20 |
| TP26 | OpEm | 29.22 | 21.93 |
| TP27 | U500_sec | 66.31 | 11.39 |
| TP28 | U0_sec | 80.66 | 6.43 |
| TP30 | U500_tr_sec | 34.56 | 6.56 |
| TP31 | U12_p | 40.78 | 44.66 |
| TP32 | U0 | 15.13 | 16.59 |
| TP33 | U0 | 35.06 | 44.66 |
| TP34 | U5_sec | 66.69 | 33.48 |
| TP35 | Isns | 12.08 | 6.70 |
| TP36 | Src | 5.60 | 8.97 |
| TP101 | Net-(TP101-Pad1) | 17.92 | -1.44 |
| TP102 | Net-(TP101-Pad1) | 20.46 | -1.44 |
| TP103 | Net-(TP101-Pad1) | 23.00 | -1.44 |
| TP104 | Net-(TP101-Pad1) | 25.54 | -1.44 |
| TP105 | Net-(TP101-Pad1) | 28.08 | -1.44 |
| TP106 | Net-(TP101-Pad1) | 30.62 | -1.44 |
| TP107 | Net-(TP107-Pad1) | 17.92 | -3.98 |
| TP108 | Net-(TP107-Pad1) | 20.46 | -3.98 |
| TP109 | Net-(TP107-Pad1) | 23.00 | -3.98 |
| TP110 | Net-(TP107-Pad1) | 25.54 | -3.98 |
| TP111 | Net-(TP107-Pad1) | 28.08 | -3.98 |
| TP112 | Net-(TP107-Pad1) | 30.62 | -3.98 |
| TP113 | Net-(TP113-Pad1) | 17.92 | -6.52 |
| TP114 | Net-(TP113-Pad1) | 20.46 | -6.52 |
| TP115 | Net-(TP113-Pad1) | 23.00 | -6.52 |
| TP116 | Net-(TP113-Pad1) | 25.54 | -6.52 |
| TP117 | Net-(TP113-Pad1) | 28.08 | -6.52 |
| TP118 | Net-(TP113-Pad1) | 30.62 | -6.52 |

All dimensions are in millimeters unless otherwise specified.

Y
X




| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
|---|---|---|---|---|
| | Board Name: **Step-Up** | | Project Name: **Isolated step-up 12:450 V** | |
| Sheet Title: Top Test Points (Scale 1:1) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: A4 | Sheet: 5 of 7 |

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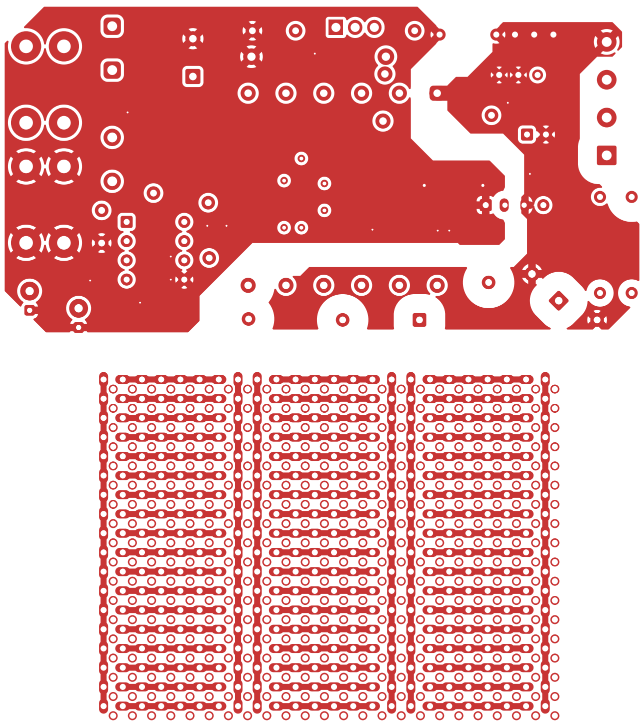
# Step-Up Fabrication Document

**F.Cu (Scale 1:1)**



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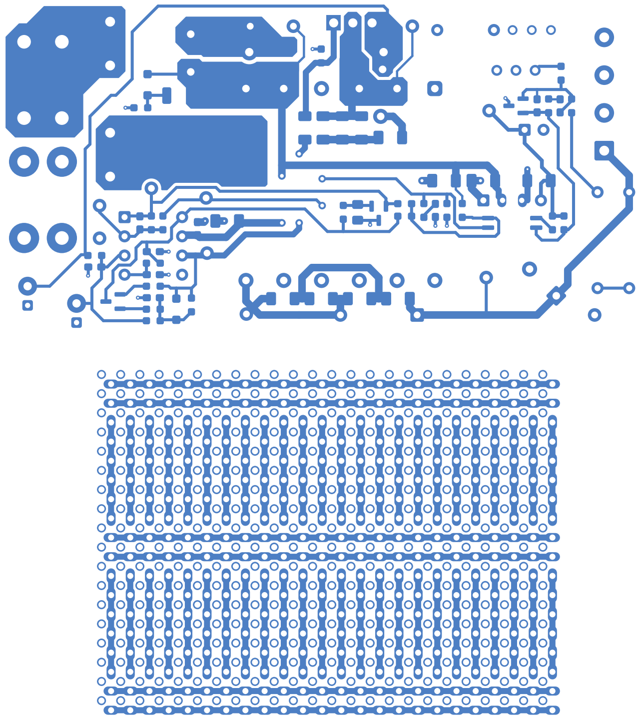
| | | | | |
|---|---|---|---|---|
| Comments: | Company: NA | | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: **Step-Up** | | Project Name: **Isolated step-up 12:450 V** | |
| Sheet Title: F.Cu (Scale 1:1) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | | Reviewer: FR | Size: **A4** | Sheet: **6** of **7** |

# Step-Up Fabrication Document

## B.Cu (Scale 1:1)



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| | | | |
|---|---|---|---|
| Comments: | Company: NA | Variant: PRELIMINARY | Git Hash: dc98ce0 |
| | Board Name: Step-Up | Project Name: Isolated step-up 12:450 V | |
| Sheet Title: B.Cu (Scale 1:1) | File Name: Isolated_12to450V.kicad_pcb | Designer: FR | Date: 2026-02-07 |
| | | | Revision: 2.0.0+ (Unreleased) |
| Sheet Path: | Reviewer: FR | Size: A4 | Sheet: 7 of 7 |